15 Vector Fields

Problem 15.1: The Gaussian Integral

Evaluate

$$\int_0^{2\pi}\int_0^{\infty} e^{-r^2} r\,\mathrm{d}r\,\mathrm{d}\theta, \qquad I=\int_{-\infty}^{\infty} e^{-x^2}\,\mathrm{d}x.$$

Solution 15.1

With $u=r^2$, we have $\mathrm{d}u=2r\,\mathrm{d}r$. Therefore,

$$\int_0^{2\pi}\int_0^{\infty} e^{-r^2} r\,\mathrm{d}r\,\mathrm{d}\theta = 2\pi\left(\frac{1}{2}\int_0^{\infty} e^{-u}\,\mathrm{d}u\right)=\pi.$$

To evaluate I, square the integral and interpret the product as an integral over the whole plane:

$$\begin{aligned} I^2 &= \left(\int_{-\infty}^{\infty} e^{-x^2}\,\mathrm{d}x\right)\left(\int_{-\infty}^{\infty} e^{-y^2}\,\mathrm{d}y\right) \\ &= \iint_{\mathbb{R}^2} e^{-(x^2+y^2)}\,\mathrm{d}A \\ &= \int_0^{2\pi}\int_0^{\infty} e^{-r^2} r\,\mathrm{d}r\,\mathrm{d}\theta = \pi. \end{aligned}$$

Since $I>0$, we obtain

$$\boxed{\int_{-\infty}^{\infty} e^{-x^2}\,\mathrm{d}x=\sqrt{\pi}.}$$

15.1 Vector Fields and Their Plots

Definition 15.1: Vector Field

A **vector field** on a domain $\mathcal{D} \subseteq \mathbb{R}^n$ is a function

$$\vec{F} : \mathcal{D} \longrightarrow \mathbb{R}^n$$

that assigns a vector $\vec{F}(P)$ to each point $P \in \mathcal{D}$. In a plot, the vector is drawn with its tail at P.

In two and three dimensions, respectively, we write

$$\vec{F}(x, y) = \langle F_1(x, y), F_2(x, y) \rangle ,$$
$$\vec{F}(x, y, z) = \langle F_1(x, y, z), F_2(x, y, z), F_3(x, y, z) \rangle .$$

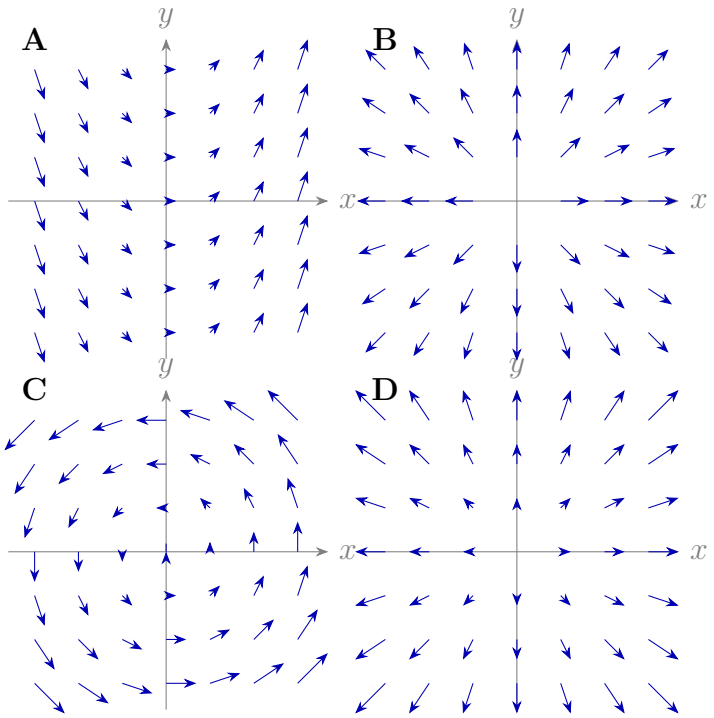
The scalar-valued functions F_1, F_2, F_3 are called the **components** of $\vec{F}$.

To identify a vector field from a plot, compare both the **direction** and the **magnitude** of its vectors. Check where the field vanishes or is undefined, and how it changes as the point moves along a coordinate direction.

Example 15.1 (Matching Vector Fields in Two Dimensions)**.**

To distinguish the following fields, examine how their directions and magnitudes change across the plots:

$$\vec{F}_1(x,y) = \langle 1, x \rangle, \qquad \vec{F}_2(x,y) = \langle 2x, 2y \rangle, \qquad \vec{F}_3(x,y) = \frac{\langle x, y \rangle}{\sqrt{x^2+y^2}}.$$



Arrows are scaled by a constant within each panel for readability.

1. $\vec{F}_1$ matches **A**. Every vector points right, with vertical component x: downward for $x < 0$, horizontal for $x = 0$, and upward for $x > 0$. The field is independent of y.

2. $\vec{F}_2$ matches **D**. Vectors point radially outward with magnitude $2\sqrt{x^2+y^2}$. They grow longer farther from the origin, where the field is $\vec{0}$.

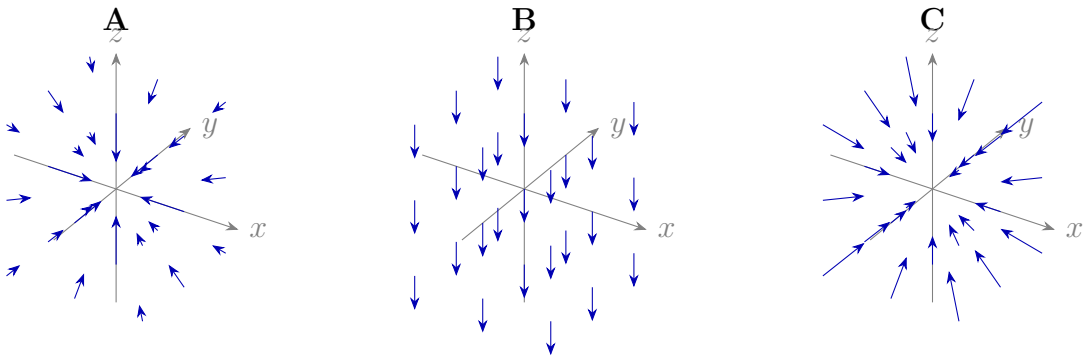
3. $\vec{F}_3$ matches **B**. Vectors point radially outward with constant magnitude 1. The field is undefined at the origin.

Plot **C** shows counterclockwise rotation, as in $\langle -y, x \rangle$, so it does not match any of the given fields.

Example 15.2 (Matching Vector Fields in Three Dimensions)**.**

To distinguish the following fields, examine how their directions and magnitudes change across the plots:

$$\vec{F}_1(x,y,z) = \langle 0, 0, -1 \rangle ,$$
$$\vec{F}_2(x,y,z) = -x\,\vec{i} - y\,\vec{j} - z\,\vec{k},$$
$$\vec{F}_3(x,y,z) = \frac{\langle -x, -y, -z \rangle}{(x^2+y^2+z^2)^{3/2}}.$$



Arrows are scaled by a constant within each panel for readability.

Let $r = \sqrt{x^2+y^2+z^2}$ denote the distance from the origin.

1. $\vec{F}_1$ matches **B**: all vectors point downward with constant magnitude 1.

2. $\vec{F}_2$ matches **C**: vectors point toward the origin and have magnitude r. They become shorter near the origin, where the field is $\vec{0}$.

3. $\vec{F}_3$ matches **A**: vectors point toward the origin and have magnitude

$$\left\| \vec{F}_3 \right\| = \frac{r}{r^3} = \frac{1}{r^2}, \qquad r > 0.$$

   They become longer near the origin and shorter farther away. The field is undefined at the origin.

15.2 Operations on Vector Fields

Definition 15.2: The Del Operator

In three dimensions, the **del operator** is

$$\nabla = \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle.$$

In two dimensions, omit the z-component.

15.2.1 Divergence

Definition 15.3: Divergence

For a differentiable vector field $\vec{F} = \langle F_1, F_2, F_3 \rangle$, its **divergence** is the scalar-valued function

$$\operatorname{div} \vec{F} = \nabla \cdot \vec{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}.$$

In two dimensions, omit the term $\frac{\partial F_3}{\partial z}$.

Divergence measures local net outward flow: positive divergence indicates a local source, negative divergence indicates a local sink, and zero divergence indicates no net local outflow.

Example 15.3 (Computing Divergence at a Point)**.**

Consider $\vec{F} = \langle 2xy, yz, e^z \rangle$. To find its divergence at $(1, 0, 0)$, first compute the divergence as a function of position.

Differentiate each component with respect to its corresponding variable:

$$\begin{aligned} \nabla \cdot \vec{F} &= \frac{\partial (2xy)}{\partial x} + \frac{\partial (yz)}{\partial y} + \frac{\partial e^z}{\partial z} \\ &= 2y + z + e^z. \end{aligned}$$

Hence,

$$(\nabla \cdot \vec{F})(1, 0, 0) = 0 + 0 + 1 = \boxed{1}.$$

15.2.2 Curl

Definition 15.4: Curl

For a differentiable vector field $\vec{F} = \langle F_1, F_2, F_3 \rangle$ in $\mathbb{R}^3$, its **curl** is the vector field

$$\begin{aligned}\operatorname{curl} \vec{F} = \nabla \times \vec{F} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix} \\ &= \left\langle \frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}, \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}, \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right\rangle.\end{aligned}$$

Curl is **vector-valued** and describes local rotation, with direction given by the right-hand rule.

Example 15.4 (Curl of a Rotational Field)**.**

Consider the rotational field $\vec{F} = \langle -y, x, 0 \rangle$. Its curl describes both the direction and the strength of its local rotation.

Using the component formula,

$$\begin{aligned}\nabla \times \vec{F} &= \left\langle 0 - 0, 0 - 0, \frac{\partial x}{\partial x} - \frac{\partial(-y)}{\partial y} \right\rangle \\ &= \langle 0, 0, 1 - (-1) \rangle = \boxed{\langle 0, 0, 2 \rangle}.\end{aligned}$$

Counterclockwise rotation in the xy-plane gives curl in the positive z-direction.

15.3 Conservative Vector Fields and Potential Functions

Definition 15.5: Conservative Vector Field

A vector field $\vec{F}$ is **conservative**, or a **gradient vector field**, if there exists a scalar-valued function f such that

$$\vec{F} = \nabla f.$$

The function f is called a **potential function** for $\vec{F}$.